Planck length: Size (10^{-33} centimeters) below which the conflict between quantum mechanics and general relativity becomes manifest; size below which conventional notion of space breaks down.

Planck mass: Mass (10^{-5} grams, mass of a grain of dust; ten billion billion times the proton mass); typical mass of a vibrating string.

Planck time: Time (10^{-43} seconds) it takes light to traverse one Planck length; time interval below which conventional notion of time breaks down.

phase transition: Qualitative change in a physical system when its temperature is varied through a sufficiently wide range.

photon: Messenger particle of the electromagnetic force; a “bundle” of light.

potential energy: Energy stored in a field or object.

potential energy bowl: Shape describing the energy a field contains for a given field value; technically called the field’s potential energy.

probability wave: Wave in quantum mechanics that encodes the probability that a particle will be found at a given location.

quantum chromodynamics: Quantum mechanical theory of the strong nuclear force.

quantum fluctuations, quantum jitters: The unavoidable, rapid variations in the value of a field on small scales, arising from quantum uncertainty.

quantum measurement problem: Problem of explaining how the myriad possibilities encoded in a probability wave give way to a single outcome when measured.

quantum mechanics: Theory, developed in the 1920s and 1930s, for describing the realm of atoms and subatomic particles.

quarks: Elementary particles subject to the strong nuclear force; there are six varieties (up, down, strange, charm, top, bottom).

relationist: Perspective holding that all motion is relative and space is not absolute.

rotational invariance, rotational symmetry: Characteristic of a physical system, or of a theoretical law, of being unaffected by a rotation.

second law of thermodynamics: Law that says that, on average, the entropy of a physical system will tend to rise from any given moment.

spacetime: The union of space and time first articulated by special relativity.

special relativity: Einstein’s theory in which space and time are not individually absolute, but instead depend upon the relative motion between distinct observers. spin: Quantum mechanical property of elementary particles in which, somewhat like a top, they undergo rotational motion (they have intrinsic angular momentum).

spontaneous symmetry breaking: Technical name for the formation of a Higgs ocean; process by which a previously manifest symmetry is hidden or spoiled.